

PROGRAM 1: INSURANCE DATABASE NAME: Adithi P Jain USN: 1WN25CS016

The insurance database must maintain information about drivers, the cars they own, the accidents

reported, and the participation of each driver and car in those accidents. Each driver in the system is uniquely identified by a driver ID, along with their name and address, and each car is uniquely identified by its registration number together with details such as model and manufacturing year. The system must allow storing ownership information that links a driver to one or more cars, while also allowing a car to be linked to one or more drivers if shared ownership occurs; duplicate ownership records for the same driver and car must not exist.

Accident information must be stored using a unique report number assigned to each accident, along with the date on which the accident occurred and the location where it happened. Every accident reported in the system must have at least one participating driver and car, and this participation is recorded by linking the driver, the involved car, and the accident report together with the corresponding damage amount for that particular involvement. A participation record must reference an existing driver, an existing car, and an existing accident, and no two participation entries may repeat the same combination of driver, car, and accident report. The database must ensure that damage amounts are non-negative, accident dates are valid calendar

dates, and car manufacturing years fall within reasonable limits. It must also preserve referential

integrity so that ownership or participation entries cannot exist without valid driver, car, and accident information already present in the system. Deletion policies must prevent removal of drivers or cars that appear in past accident participation records unless historical consistency is

preserved through controlled deletion rules or archival mechanisms. The system should maintain accurate links between drivers, cars, and accidents at all times, ensuring reliable retrieval of ownership histories, accident histories, and damage information for administrative, legal, and

insurance-related purpose

Consider the Insurance database given below. The data types are specified.

PERSON (driver_id: String, name: String, address: String)

CAR (reg_num: String, model: String, year: int)

ACCIDENT (report_num: int, accident_date: date, location: String)

OWNS (driver_id: String, reg_num: String)

PARTICIPATED (driver_id: String, reg_num: String, report_num: int, damage_amount: int)

```
CREATE DATABASE INSURANCE_DATABASE;
```

```
CREATE TABLE PERSON (  
    driver_id VARCHAR(10) PRIMARY KEY,  
    name  VARCHAR(50) NOT NULL,  
    address VARCHAR(100)  
);
```

```
CREATE TABLE CAR (  
    reg_num VARCHAR(15) PRIMARY KEY,  
    model VARCHAR(30) NOT NULL,  
    year INT NOT NULL,  
  
);
```

```
CREATE TABLE ACCIDENT (  
    report_num INT PRIMARY KEY,  
    accident_date DATE NOT NULL,  
    location VARCHAR(100)  
);
```

```
CREATE TABLE OWNS (  
    driver_id VARCHAR(10),  
    reg_num VARCHAR(15),  
    FOREIGN KEY (driver_id) REFERENCES PERSON(driver_id)  
    FOREIGN KEY (reg_num) REFERENCES CAR(reg_num)  
  
);
```

```
CREATE TABLE PARTICIPATED (  
    driver_id VARCHAR(10),  
    reg_num VARCHAR(15),  
    report_num INT,  
    damage_amount INT  
    FOREIGN KEY (driver_id) REFERENCES PERSON(driver_id)
```

```
FOREIGN KEY (reg_num) REFERENCES CAR(reg_num)
FOREIGN KEY (report_num) REFERENCES ACCIDENT(report_num)
CONSTRAINT chk_damage CHECK (damage_amount >= 0)
);
```

INSERT INTO PERSON VALUES

```
('A01', 'Richard', 'Srinivas nagar'),
('A02', 'Pradeep', 'Rajaji nagar'),
('A03', 'Smith', 'Ashok nagar'),
('A04', 'Venu', 'N R Colony'),
('A05', 'Jhon', 'Hanumanth nagar');
```

INSERT INTO CAR VALUES

```
('KA052250', 'Indica', 1990),
('KA031181', 'Lancer', 1957),
('KA095477', 'Toyota', 1998),
('KA053408', 'Honda', 2008),
('KA041702', 'Audi', 2005);
```

INSERT INTO ACCIDENT VALUES

```
(11, '2003-01-01', 'Mysore Road'),
(12, '2004-02-02', 'South end Circle'),
(13, '2003-01-21', 'Bull temple Road'),
(14, '2008-02-17', 'Mysore Road'),
(15, '2005-03-04', 'Kanakpura Road')
```

INSERT INTO OWNS VALUES

```
('A01', 'KA052250'),
('A02', 'KA053408'),
('A03', 'KA031181'),
('A04', 'KA095477'),
('A05', 'KA041702');
```

INSERT INTO PARTICIPATED VALUES

('A01', 'KA052250', 11, 10000),

('A02', 'KA053408', 12, 50000),

('A03', 'KA095477', 13, 25000),

('A04', 'KA031181', 14, 3000),

('A05', 'KA041702', 15, 5000);

DESCRIPTION OF TABLE:

1) PERSON

1 • desc person;

res

Field	Type	Null	Key	Default	Extra
DRIVER_ID	varchar(50)	NO	PRI	NULL	
NAME	varchar(50)	NO		NULL	
ADDRESS	varchar(50)	YES		NULL	

2) OWNS

1 • desc owns;

tures

1 2

Field	Type	Null	Key	Default	Extra
DRIVER_ID	varchar(50)	NO	PRI	NULL	
REG_NO	varchar(50)	NO	PRI	NULL	

3) ACCIDENT

1 • desc accident;

Result Grid

Field	Type	Null	Key	Default	Extra
REPORT_NUM	int	NO	PRI	NULL	
ACCIDENT_DATE	date	NO		NULL	
LOCATION	varchar(50)	NO		NULL	

4) PARTICIPATED

1 • desc participated;

Result Grid

Field	Type	Null	Key	Default	Extra
DRIVER_ID	varchar(50)	YES	MUL	NULL	
REG_NO	varchar(50)	YES	MUL	NULL	
REPORT_NUM	int	YES	MUL	NULL	
DAMAGE_AMT	int	YES		NULL	

5) CAR

1 • desc car;

Result Grid

Field	Type	Null	Key	Default	Extra
REG_NO	varchar(50)	NO	PRI	NULL	
MODEL	varchar(50)	NO		NULL	
YEAR	int	YES		NULL	

INSERTION OF VALUES:

```
5 • select * FROM participated ;
```

```
6
```

```
7
```

Result Grid				
Filter Rows:				
Export: 				
	DRIVER_ID	REG_NO	REPORT_NUM	DAMAGE_AMT
▶	A01	KA052250	11	10000
	A02	KA053408	12	50000
	A03	KA095477	13	25000
	A04	KA031181	14	3000
	A05	KA041702	15	5000

PARTICIPATED

```

1
2 • select * FROM person ;
3 • select * FROM accident ;
4 • select * FROM car ;
5 • select * FROM participated ;
6
7

```

Result Grid

Filter Rows:

Edit:

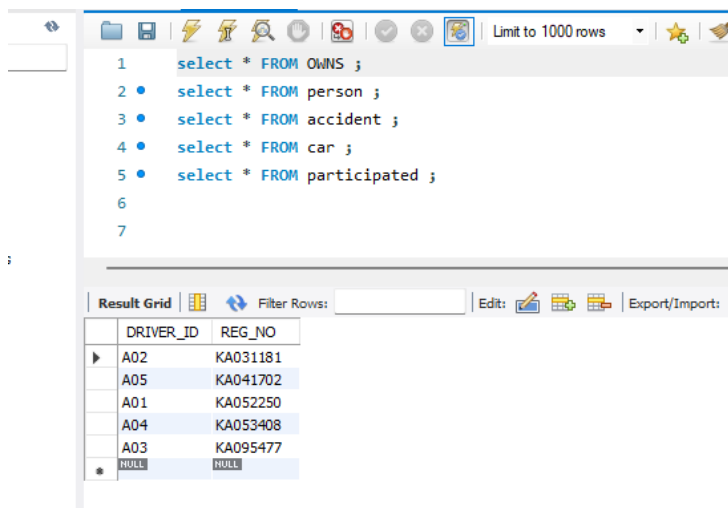
	DRIVER_ID	NAME	ADRESS
▶	A01	RICHARD	SRINIVAS NAGAR
	A02	PRADEEP	RAJAJI NAGAR
	A03	SMITH	ASHOK NAGAR
	A04	VENU	N R COLONY
	A05	JHON	HANUMANTH NAGAR
*	NULL	NULL	NULL

PERSON

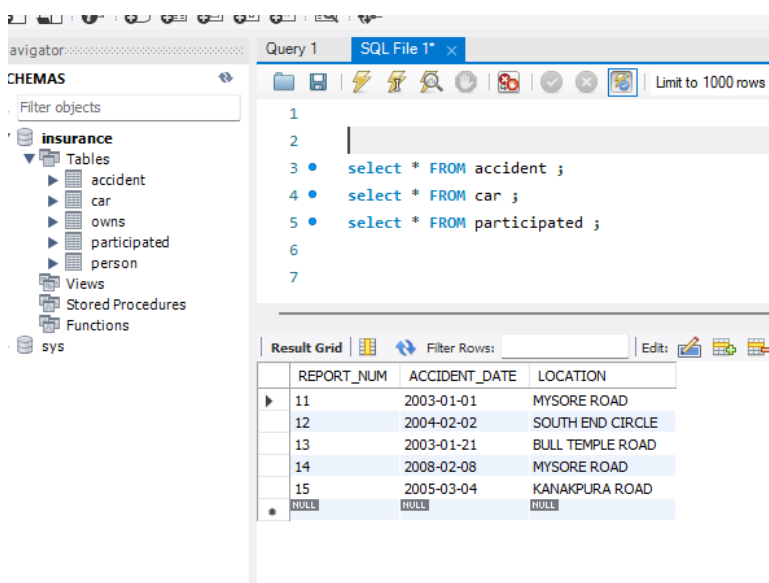
2	
3	
4 •	select * FROM car ;
5 •	select * FROM participated ;
6	
7	

Result Grid			
Filter Rows:			
Edit:			
	REG_NO	MODEL	YEAR
▶	KA031181	LANCER	1957
	KA041702	AUDI	2005
	KA052250	INDICA	1990
	KA053408	HONDA	2008
	KA095477	TOYOTA	1998
*	NULL	NULL	NULL

CAR



OWNS

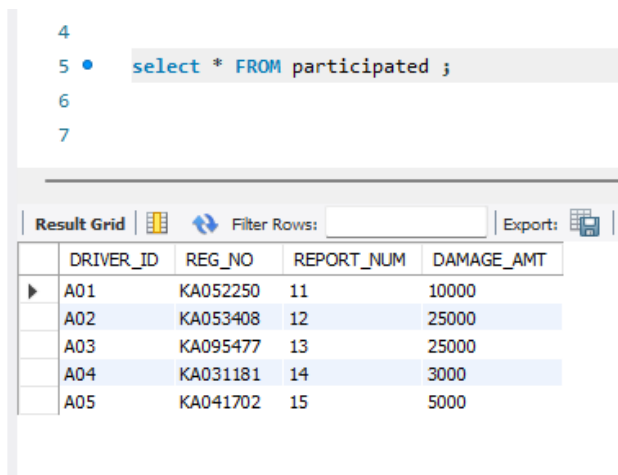


ACCIDENT

SCREENSHOTS: EXECUTION OF QUERY

Update the damage amount to 25000 for the car with a specific reg_num (example: 'K

A053408') for which the accident report number was 12.



Add a new accident to the database.

Display Accident date and location

```
15 • update participated set damage_amt = 25000 where reg_no = "KA053408";
16 • insert into accident values(16,"2026-01-01","RV ROAD");
17 • select accident_date,location from accident;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
accident_date	location			
2003-01-01	MYSORE ROAD			
2004-02-02	SOUTH END CIRCLE			
2003-01-21	BULL TEMPLE ROAD			
2008-02-08	MYSORE ROAD			
2005-03-04	KANAKPURA ROAD			
2026-01-01	RV ROAD			

Display driver id who did the accident damage greater than or equal to Rs.25000

```
14
15 • update participated set damage_amt = 25000 where reg_no = "KA053408";
16 • insert into accident values(16,"2026-01-01","RV ROAD");
17 • select accident_date,location from accident;
18 • select driver_id from participated where damage_amt>=25000;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
driver_id				
A02				
A03				